Generalizing Few-Shot Classification of Whole-Genome Doubling Across Cancers

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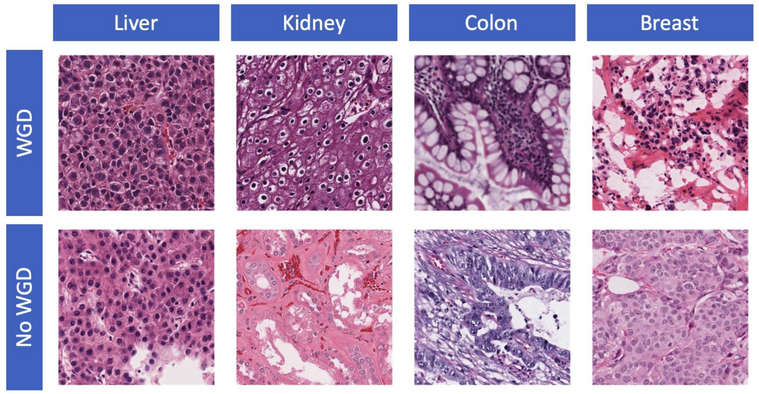
Summary

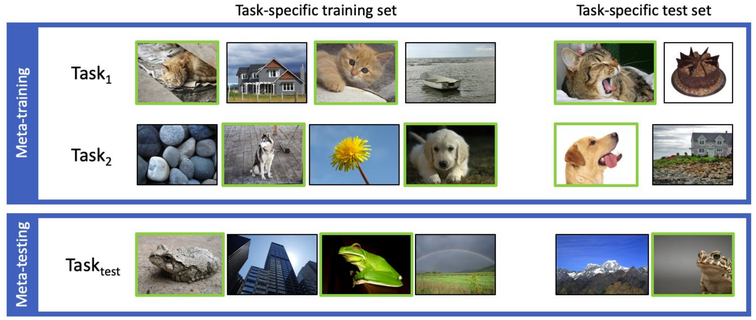
In this work, we demonstrate that meta-learning enables signal extraction from medical imaging data mired in tissue site-specific idiosyncrasies. Unlabeled data is often abundant in healthcare settings because label acquisition is expensive. The meta-learning training regime enables fast learning with only a handful of labeled examples. Using digitized histopathology slide images of primary tumor biopsies, we train a deep neural network model end-to-end to accurately generalize few-shot classification of whole-genome doubling across 17 cancer types. This approach outperforms a traditional neural network classifier and quickly generalizes to held-out cancer types and batch effects.

Background/Motivation



- Whole-genome doubling (WGD) is a genomic aberration of twice the normal number of chromosomes and is associated with advanced metastasis and overall poor prognosis.
- While WGD events occur across cancer types, it is a nontrivial amount of time and money to measure WGD.
- Increasingly, efforts are being made to determine genomic characteristics of a patient's tumor from readily-available H&E-stained whole-slide images for individual cancer types.
- However, training one classifier across cancer types is made difficult by tissue-specific heterogeneity and batch effects (e.g., systematic differences in image resolution and brightness).

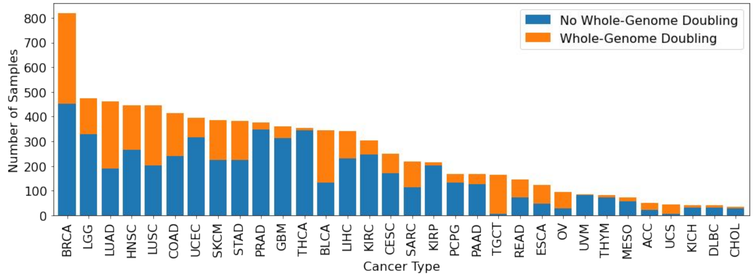
Methods



- To address the task of generalizing WGD classification across cancer types, we apply a meta-learning framework.
- Under meta-learning, the model is trained on multiple tasks (meta-training).
- When the model is deployed, it is presented with a small set of labeled examples and quickly bootstraps a task-specific classifier (meta-testing).
- In contrast, under standard supervised learning, the model remains static at deployment time.

Data

- Our dataset is comprised of 17 cancer types from The Cancer Genome Atlas (TCGA), totaling nearly 3,500 H&E-stained whole-slide images, 42% of which are WGD-positive.
- We meta-train on eight cancer types, meta-validate on two cancer types, and meta-test on remaining seven held-out cancer types.

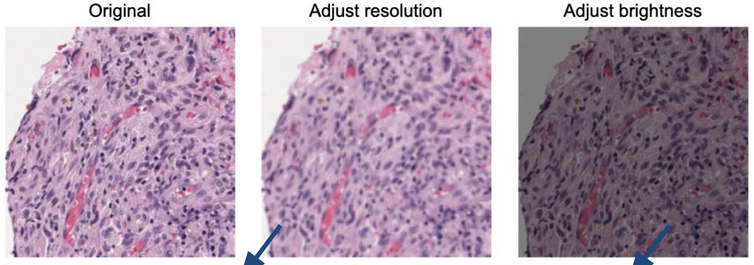


Results

- The meta-learning algorithm (MAML) outperforms the standard machine learning training algorithm (CNN).

	CNN	MAML
ACC	0.6873 ± 0.0540	**0.6988 ± 0.0581**
CHOL	**0.6890 ± 0.0532**	0.6845 ± 0.0643
KICH	0.6928 ± 0.0312	**0.7022 ± 0.0303**
KIRC	0.6611 ± 0.0609	**0.6843 ± 0.1018**
OV	0.6950 ± 0.0393	**0.7020 ± 0.0435**
UCEC	**0.7000 ± 0.0602**	0.6859 ± 0.0816
UCS	0.6846 ± 0.0387	**0.6908 ± 0.0667**
META-TEST	0.6888 ± 0.0506	**0.6944 ± 0.0773**

- When we apply a 50% reduction in either resolution or brightness to the held-out meta-test images, we find that MAML continues to exhibit performance superior to that of the CNN.



	CNN	MAML
ACC	0.6316 ± 0.1011	**0.6930 ± 0.0689**
CHOL	**0.6733 ± 0.0570**	0.6730 ± 0.0548
KICH	**0.7081 ± 0.0261**	0.6941 ± 0.0285
KIRC	0.6269 ± 0.0805	**0.6401 ± 0.0855**
OV	0.7097 ± 0.0728	**0.7136 ± 0.0751**
UCEC	0.6794 ± 0.0496	**0.6877 ± 0.0569**
UCS	**0.6697 ± 0.0330**	0.6649 ± 0.0886
META-TEST	0.6713 ± 0.0716	**0.6809 ± 0.0717**

	CNN	MAML
ACC	0.6742 ± 0.0801	**0.7062 ± 0.0655**
CHOL	0.6819 ± 0.0475	**0.7147 ± 0.0388**
KICH	**0.7101 ± 0.0216**	0.7079 ± 0.0335
KIRC	0.6670 ± 0.0795	**0.7060 ± 0.0825**
OV	0.6989 ± 0.0774	**0.7000 ± 0.0952**
UCEC	0.6899 ± 0.0312	**0.6961 ± 0.0379**
UCS	**0.6900 ± 0.0495**	0.6837 ± 0.0874
META-TEST	0.6884 ± 0.0620	**0.6973 ± 0.0704**

Conclusion

Accounting for variations between cancer types is made possible by fast learning on only a handful of labeled images. These results demonstrate that meta-learning enables real-time identification of cancer phenotypes that are too often costly and inefficient to obtain.

Acknowledgements

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